

BRAC University
MAT-215
Exercise Sheet # 2

1. Evaluate the following Limits:

$$(i) \lim_{z \rightarrow 1+i} \frac{z^2 - z + 1 - i}{z^2 - 2z + 2} \quad (ii) \lim_{z \rightarrow 1+i} \left\{ \frac{z - 1 - i}{z^2 - 2z + 2} \right\}^2 \quad (iii) \lim_{z \rightarrow i} \frac{z^2 + 1}{z^6 + 1}.$$

2. Use the definition of neighborhood to solve the following problems:

- (i) Show that $\lim_{z \rightarrow i} 2z = 2i$
(ii) Show that if $f(z) = \frac{i\bar{z}}{2}$ then $\lim_{z \rightarrow 1} f(z) = \frac{i}{2}$.

3. If $f(z) = \frac{2z-1}{3z+2}$, prove that $\lim_{h \rightarrow 0} \frac{f(z_0 + h) - f(z_0)}{h} = \frac{7}{(3z_0 + 2)^2}$ provided $z_0 \neq -\frac{2}{3}$.

4. Let $f(z) = \frac{z^2 + 4}{z - 2i}$ if $z \neq 2i$, while $f(2i) = 3 + 4i$, is $f(z)$ continuous at $z = 2i$.

5. Find all points of discontinuity for the function $f(z) = \frac{2z-3}{z^2 + 2z + 2}$.

6. Using the definitions, find the derivative of each function at the indicated points

- (i) $f(z) = \frac{2z-i}{z+2i}$ at $z = -i$
(ii) $f(z) = 3z^{-2}$ at $z = 1+i$.

7. Evaluate the following Limits using L' Hospital's rule

$$(i) \lim_{z \rightarrow 2i} \frac{z^2 + 4}{2z^2 + (3-4i)z - 6i} \quad (ii) \lim_{z \rightarrow 0} \frac{z - \sin z}{z^3} \quad (iii) \lim_{z \rightarrow 0} \left(\frac{\sin z}{z} \right)^{\frac{1}{z^2}}$$

8. Determine which of the following functions u are harmonic. For each harmonic function find the conjugate harmonic function v and express $u + iv$ as an analytic function of z

- (i) $u = 3x^2y + 2x^2 - y^3 - 2y^2$
(ii) $u = xe^x \cos y - ye^x \sin y$
(iii) $u = e^{-x}(x \sin y - y \cos y)$.